# Labsheet 7 SECS05-AS07

# Mastery in Data Analytics and Visualizations and Career Advancement


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In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv('Mall_Customers.csv')
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

In [ ]:
```
# Check for missing values and duplicates
print("Missing values:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
df.rename(columns={
    'Annual Income (k$)': 'Annual_Income',
    'Spending Score (1-100)': 'Spending_Score'
}, inplace=True)

# Select features for clustering
X = df[['Age', 'Annual_Income', 'Spending_Score']]
```

```
Missing values:
 CustomerID                0
Gender                     0
Age                        0
Annual Income (k$)         0
Spending Score (1-100)     0
dtype: int64
Duplicates: 0
```

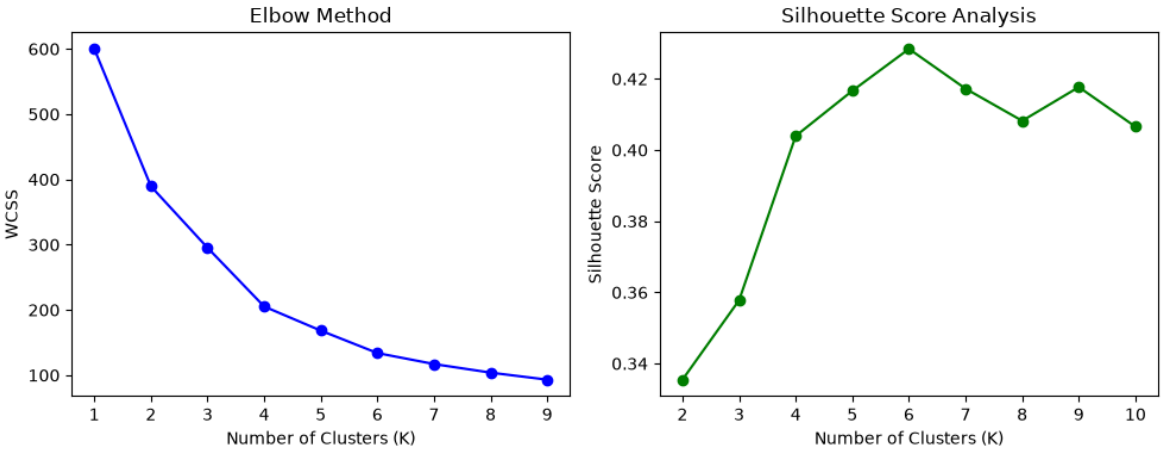
In [5]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [10]:
```
wcss = []
silhouette_scores = []
k_range = range(2, 11)
for k in k_range:
    kmeans = KMeans(n_clusters=k, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
    silhouette_scores.append(silhouette_score(X_scaled, kmeans.labels_))
fig, ax = plt.subplots(1, 2, figsize=(10, 4))
```

```
# Elbow Method
ax[0].plot(range(1, 10), [KMeans(n_clusters=i, init='k-means++', random_state=42
ax[0].set_title('Elbow Method')
ax[0].set_xlabel('Number of Clusters (K)')
ax[0].set_ylabel('WCSS')

# Silhouette Scores
ax[1].plot(k_range, silhouette_scores, marker='o', color='green')
ax[1].set_title('Silhouette Score Analysis')
ax[1].set_xlabel('Number of Clusters (K)')
ax[1].set_ylabel('Silhouette Score')

plt.tight_layout()
plt.show()
```

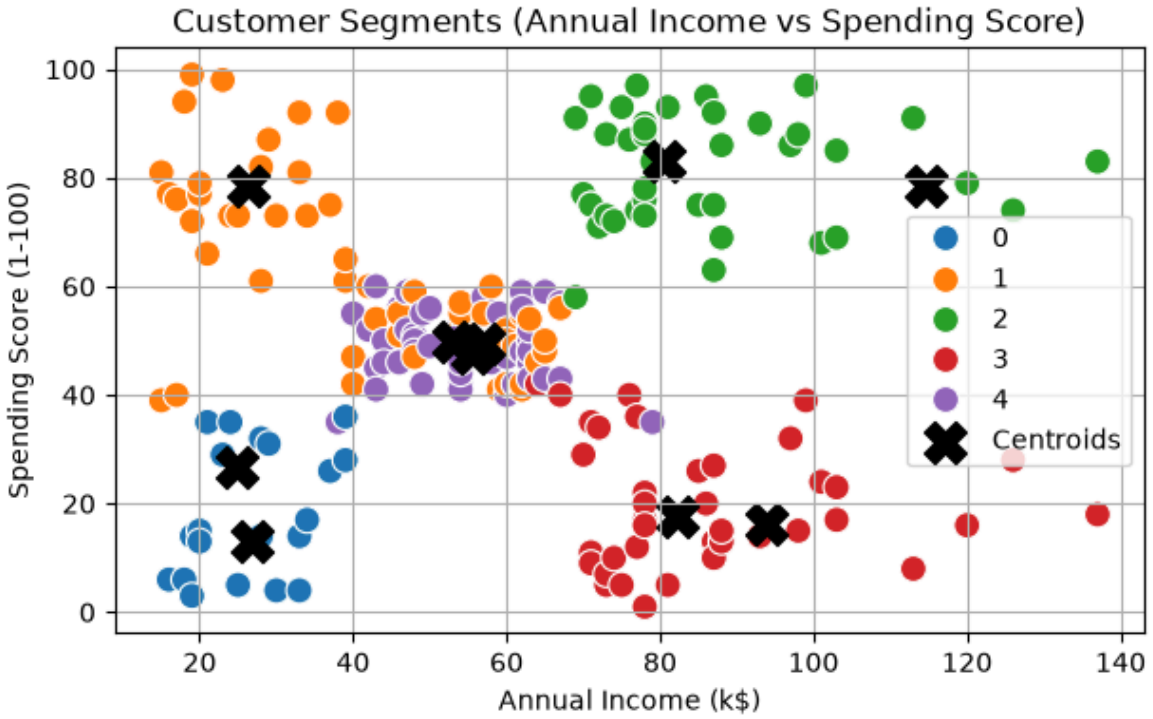



In [ ]:
```
optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42, n_init=
df['Cluster'] = kmeans.fit_predict(X_scaled)
print(df.head())
```

| | CustomerID | Gender | Age | Annual_Income | Spending_Score | Cluster |
|---|---|---|---|---|---|---|
| 0 | 1 | Male | 19 | 15 | 39 | 1 |
| 1 | 2 | Male | 21 | 15 | 81 | 1 |
| 2 | 3 | Female | 20 | 16 | 6 | 0 |
| 3 | 4 | Female | 23 | 16 | 77 | 1 |
| 4 | 5 | Female | 31 | 17 | 40 | 1 |

In [11]:
```
plt.figure(figsize=(7, 4))
sns.scatterplot(
    x='Annual_Income',
    y='Spending_Score',
    hue='Cluster',
    data=df,
    palette='tab10',
    s=100
)
centroids_unscaled = scaler.inverse_transform(kmeans.cluster_centers_)
plt.scatter(
    centroids_unscaled[:, 1],
    centroids_unscaled[:, 2],
    s=250,
    c='black',
    marker='X',
    label='Centroids'
)
```

```
plt.title('Customer Segments (Annual Income vs Spending Score)')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.grid(True)
plt.show()
```




In [9]:
```
# Segment summary statistics
cluster_summary = df.groupby('Cluster')[['Age', 'Annual_Income', 'Spending_Score
print(cluster_summary)
```

```
                Age  Annual_Income  Spending_Score
Cluster
0         46.250000      26.750000       18.350000
1         25.185185      41.092593       62.240741
2         32.875000      86.100000       81.525000
3         39.871795      86.102564       19.358974
4         55.638298      54.382979       48.851064
```